Task 1: Term Deposit Subscription Prediction (Bank Marketing)

Problem Statement

Banks conduct marketing campaigns to promote term deposit subscriptions.
The objective of this project is to build a machine learning model that predicts whether a customer will subscribe to a term deposit based on their profile and past interactions.

Objective

- Load and explore the Bank Marketing dataset
- Perform data preprocessing and encoding
- Train classification models (Logistic Regression and Random Forest)
- Evaluate models using Confusion Matrix, F1-Score, and ROC Curve
- Explain model predictions using SHAP (Explainable AI)
- Derive insights about customer behavior

```
# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
```

```
be saved outside of the current session
import kagglehub

# Download latest version

path = kagglehub.dataset_download("alexkataev/bank-marketing-data-
set")

print("Path to dataset files:", path)
```

```
/kaggle/input/datasets/alexkataev/bank-marketing-data-set/bank-
additional-names.txt
/kaggle/input/datasets/alexkataev/bank-marketing-data-set/bank-
additional-full.csv
Path to dataset files: /kaggle/input/datasets/alexkataev/bank-
marketing-data-set
```

2. Environment Setup and Data Loading

In this section, we prepare the working environment by installing the required libraries and loading the dataset.

```
# Install SHAP for model interpretability
!pip install shap -q

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import kagglehub
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, classification_report,
roc_curve, auc, f1_score
import shap

# Download dataset
path = kagglehub.dataset_download("alexkataev/bank-marketing-data-
set")
df = pd.read_csv(f"{path}/bank-additional-full.csv", sep=';')

# Quick look at the data
print("Dataset Shape:", df.shape)
df.head()
```

```
Dataset Shape: (41188, 21)
```

```
   age        job  marital    education  default housing loan
contact  \
0   56  housemaid  married     basic.4y       no      no   no
telephone
1   57   services  married  high.school  unknown      no   no
telephone
2   37   services  married  high.school       no     yes   no
telephone
3   40     admin.  married     basic.6y       no      no   no
telephone
4   56   services  married  high.school       no      no  yes
telephone

  month day_of_week  ...  campaign  pdays  previous     poutcome
emp.var.rate  \
0   may         mon  ...         1    999         0  nonexistent
1.1
1   may         mon  ...         1    999         0  nonexistent
1.1
2   may         mon  ...         1    999         0  nonexistent
1.1
3   may         mon  ...         1    999         0  nonexistent
1.1
4   may         mon  ...         1    999         0  nonexistent
1.1

   cons.price.idx  cons.conf.idx  euribor3m  nr.employed   y
0          93.994          -36.4      4.857       5191.0  no
1          93.994          -36.4      4.857       5191.0  no
2          93.994          -36.4      4.857       5191.0  no
3          93.994          -36.4      4.857       5191.0  no
4          93.994          -36.4      4.857       5191.0  no

[5 rows x 21 columns]
```

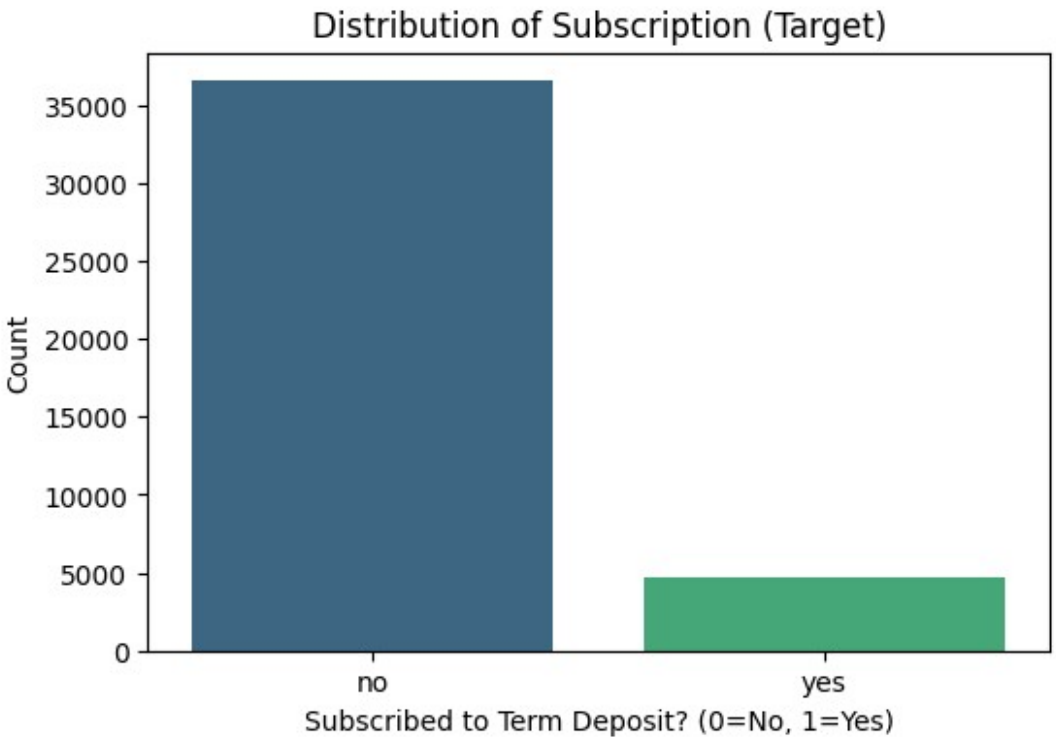
3. Exploratory Data Analysis (EDA)

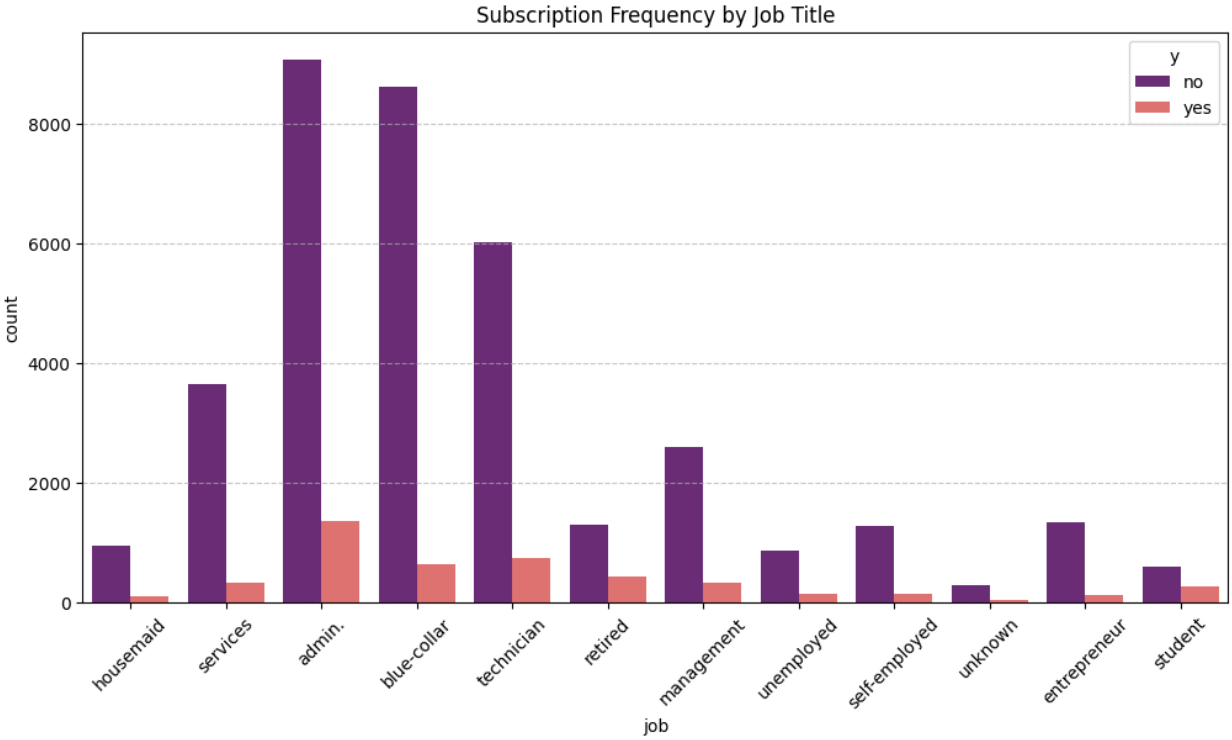
Understanding the distribution of our target variable and key features. Usually, bank marketing datasets are imbalanced, meaning there are significantly more "no" responses than "yes" responses for term deposit subscription.

```
# Check for missing values
print("Missing values per column:")
print(df.isnull().sum())

# Visualizing Target Variable Distribution
# Fixed: Added hue='y' and legend=False to resolve the FutureWarning
plt.figure(figsize=(6,4))
sns.countplot(x='y', data=df, hue='y', palette='viridis',
```

```
legend=False)
plt.title('Distribution of Subscription (Target)')
plt.xlabel('Subscribed to Term Deposit? (0=No, 1=Yes)')
plt.ylabel('Count')
plt.show()

# Visualizing Job vs Subscription
plt.figure(figsize=(12,6))
sns.countplot(x='job', hue='y', data=df, palette='magma')
plt.xticks(rotation=45)
plt.title('Subscription Frequency by Job Title')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

```
Missing values per column:
age               0
job               0
marital           0
education         0
default           0
housing           0
loan              0
contact           0
month             0
day_of_week       0
duration          0
campaign          0
pdays             0
previous          0
poutcome          0
emp.var.rate      0
cons.price.idx    0
cons.conf.idx     0
euribor3m         0
nr.employed       0
y                 0
dtype: int64
```


Distribution of Subscription (Target)
Count
35000
30000
25000
20000
15000
10000
5000
0
no
yes
Subscribed to Term Deposit? (0=No, 1=Yes)


Subscription Frequency by Job Title
y
no
yes
count
8000
6000
4000
2000
0
housemaid
services
admin.
blue-collar
technician
retired
management
unemployed
self-employed
unknown
entrepreneur
student
job

4. Data Preprocessing and Encoding

Machine learning models require numerical input. We will use Label Encoding for the target variable and One-Hot Encoding for categorical features to ensure that the model does not assume a mathematical hierarchy between categories (such as job roles).

```
# Encode the target variable
le = LabelEncoder()
df['y'] = le.fit_transform(df['y']) # 'yes' becomes 1, 'no' becomes 0

# Identify categorical columns for One-Hot Encoding
categorical_cols = df.select_dtypes(include=['object']).columns

# Create dummy variables
df_encoded = pd.get_dummies(df, columns=categorical_cols,
drop_first=True)

# Define Features (X) and Target (y)
X = df_encoded.drop('y', axis=1)
y = df_encoded['y']

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42, stratify=y)

# Feature Scaling (Crucial for Logistic Regression)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Preprocessing complete. Training set size:", X_train.shape)

Preprocessing complete. Training set size: (32950, 53)
```

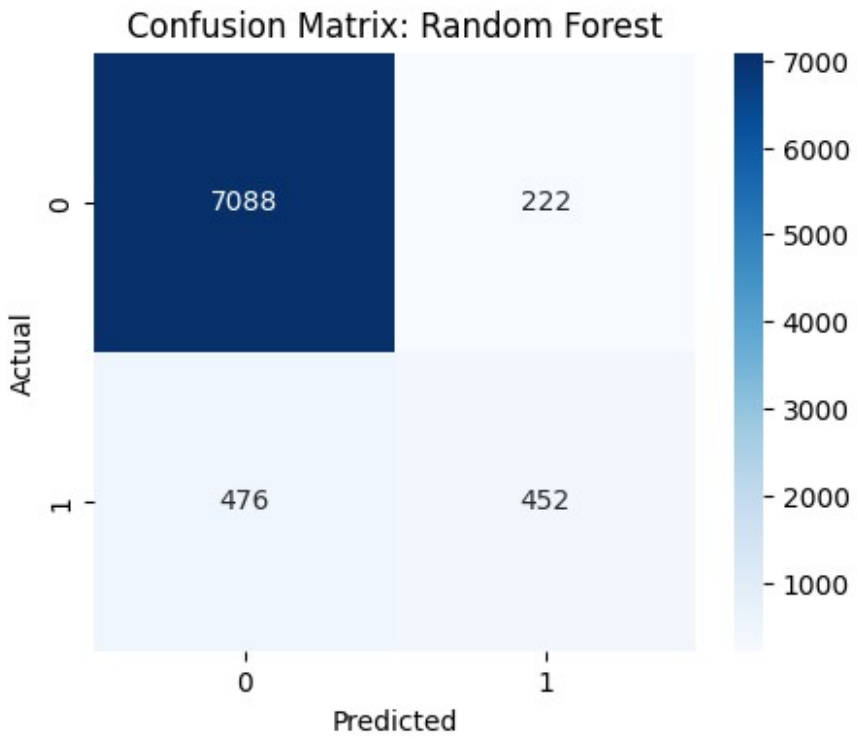
5. Model Building and Evaluation

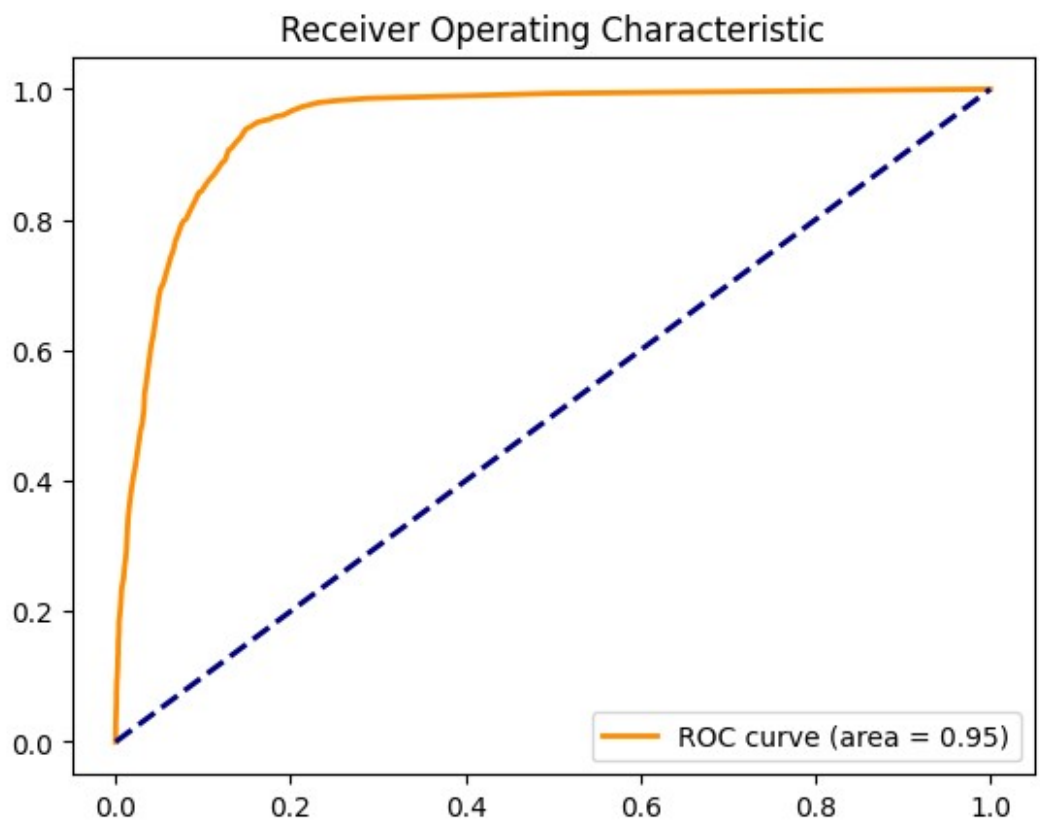
We will train a Logistic Regression model as a baseline and a Random Forest model for improved accuracy and feature importance analysis.

```
# 1. Logistic Regression
lr_model = LogisticRegression(max_iter=1000)
lr_model.fit(X_train_scaled, y_train)
lr_preds = lr_model.predict(X_test_scaled)

# 2. Random Forest
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train) # RF handles unscaled data well
rf_preds = rf_model.predict(X_test)
```

```python
# Evaluation - Classification Report
print("--- Random Forest Performance ---")
print(classification_report(y_test, rf_preds))

# Confusion Matrix Visualization
cm = confusion_matrix(y_test, rf_preds)
plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix: Random Forest')
plt.show()

# ROC Curve
fpr, tpr, _ = roc_curve(y_test, rf_model.predict_proba(X_test)[:,1])
roc_auc = auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area =
{roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.title('Receiver Operating Characteristic')
plt.legend(loc="lower right")
plt.show()
```

```
--- Random Forest Performance ---
              precision    recall  f1-score   support

           0       0.94      0.97      0.95      7310
           1       0.67      0.49      0.56       928

    accuracy                           0.92      8238
   macro avg       0.80      0.73      0.76      8238
weighted avg       0.91      0.92      0.91      8238
```


Confusion Matrix: Random Forest
7088
222
476
452
Actual
0
1
Predicted
0
1
7000
6000
5000
4000
3000
2000
1000

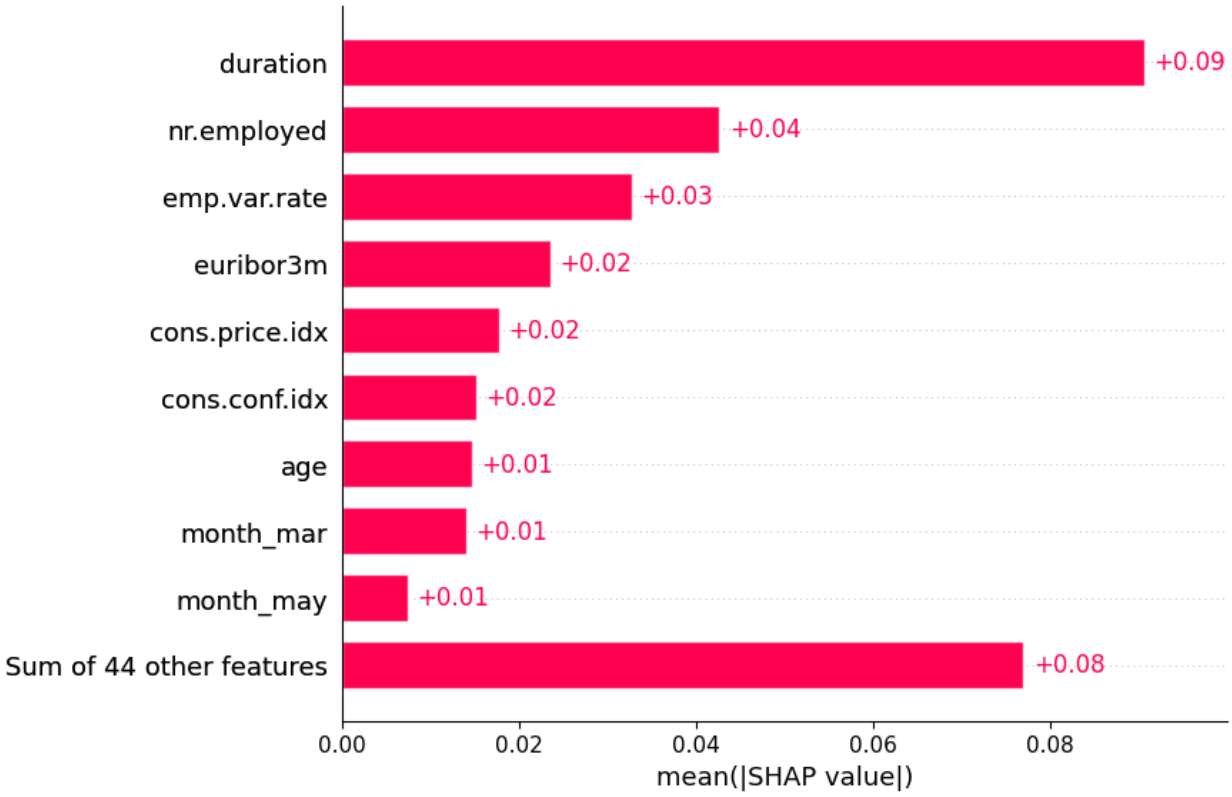

Receiver Operating Characteristic
ROC curve (area = 0.95)
0.0
0.2
0.4
0.6
0.8
1.0

6. Model Interpretability (Explainable AI - XAI)

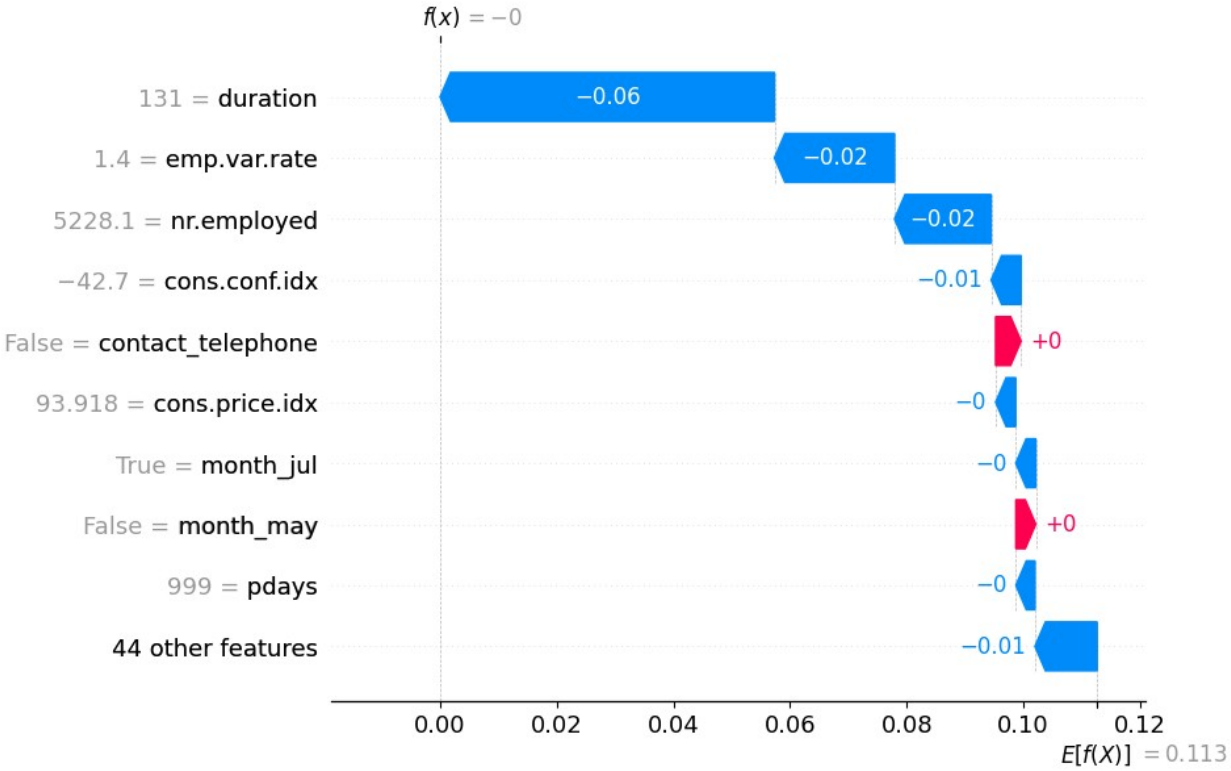
Using SHAP (SHapley Additive exPlanations) to understand why the model made specific decisions for five individual customers.

```
# 1. Initialize the SHAP Explainer
explainer = shap.TreeExplainer(rf_model)

# 2. Calculate SHAP values for a small sample
# Note: We use check_additivity=False sometimes if there are small
rounding discrepancies
X_sample = X_test.iloc[:5]
shap_values = explainer(X_sample)

# 3. Global Feature Importance Plot (Summary)
print("Global Feature Importance (SHAP Summary):")
shap.plots.bar(shap_values[:,:,1]) # Index 1 is the 'Yes' subscription
class

# 4. Individual prediction explanations (Waterfall Plot)
print("\n--- Individual Explanations for the first 5 customers ---")

for i in range(5):
    prediction_prob = rf_model.predict_proba(X_sample.iloc[[i]])[0][1]
    prediction_label = "Will Subscribe" if prediction_prob > 0.5 else
"Will Not Subscribe"

    print(f"\nCustomer Index {i} | Prediction: {prediction_label}
(Prob: {prediction_prob:.2f})")

    # The waterfall plot is the best replacement for force_plot in
static notebooks
    # .values[:,:,1] selects the SHAP values for the 'Yes' class
    shap.plots.waterfall(shap_values[i,:,1])
```

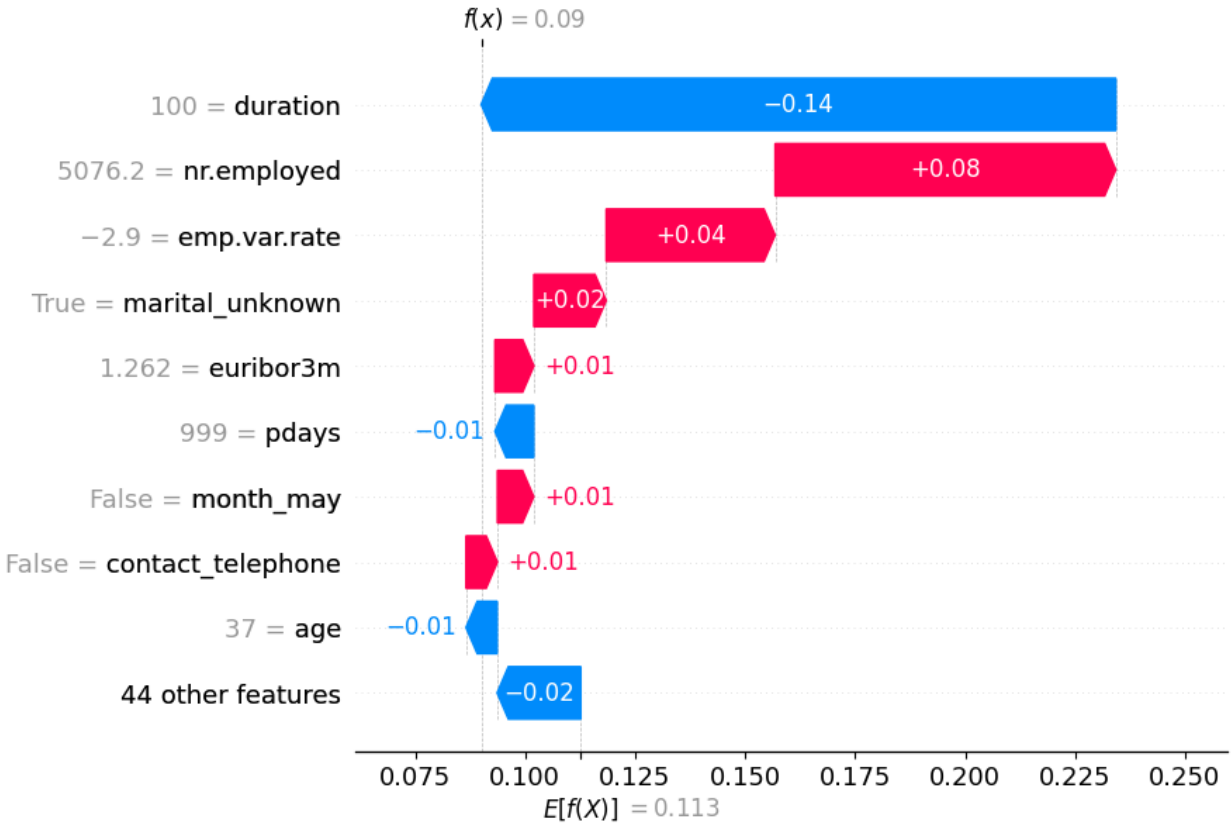
```
Global Feature Importance (SHAP Summary):
```



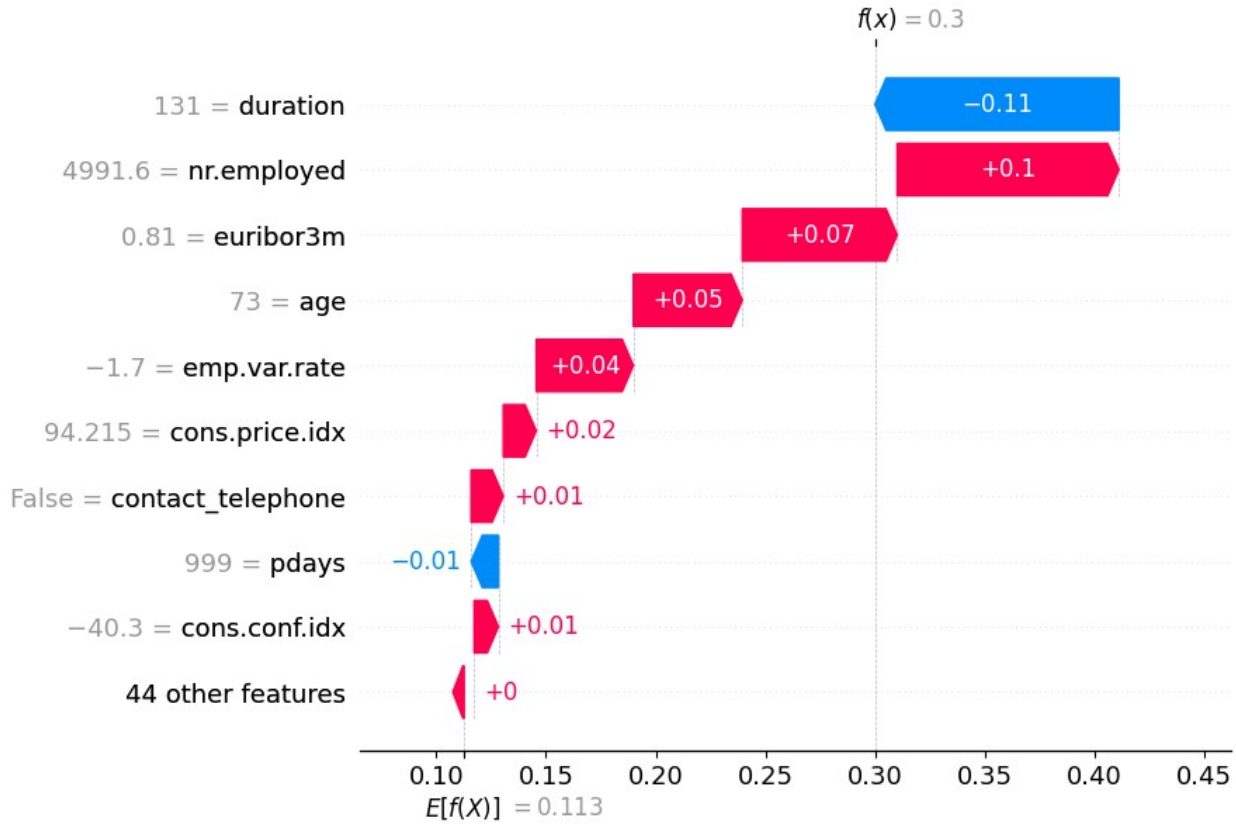

```
--- Individual Explanations for the first 5 customers ---

Customer Index 0 | Prediction: Will Not Subscribe (Prob: 0.00)
```



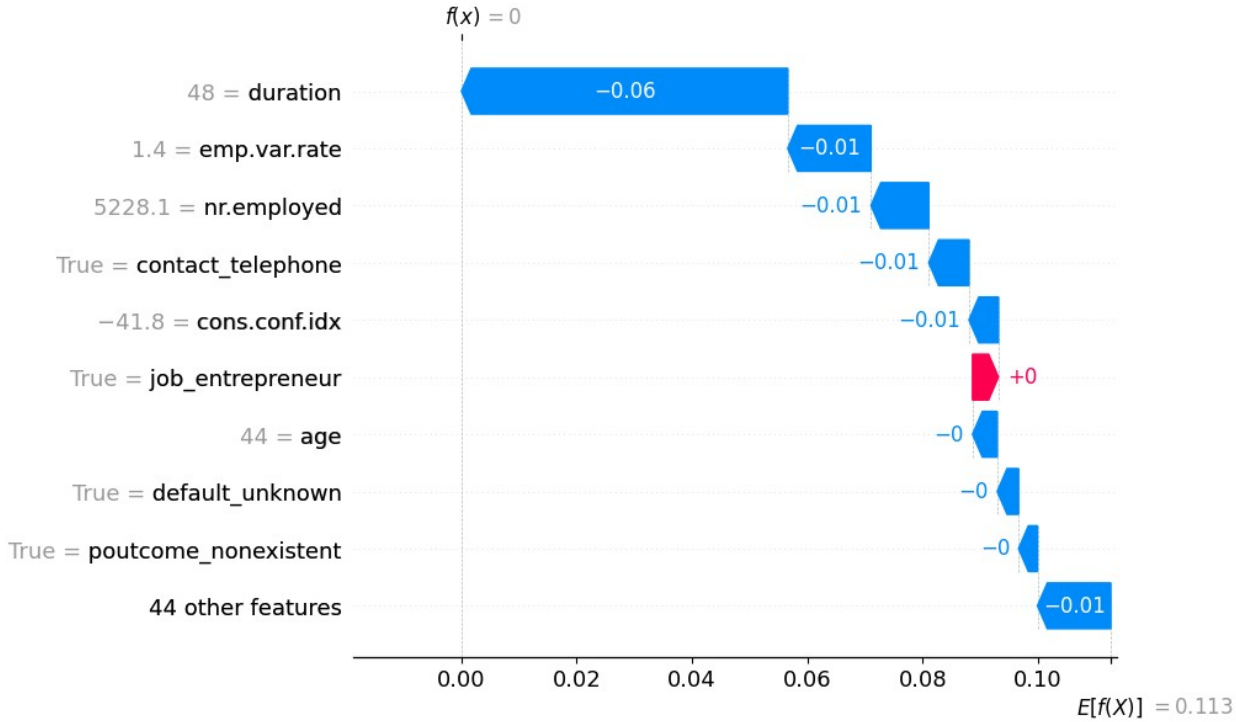

```
Customer Index 1 | Prediction: Will Not Subscribe (Prob: 0.09)
```



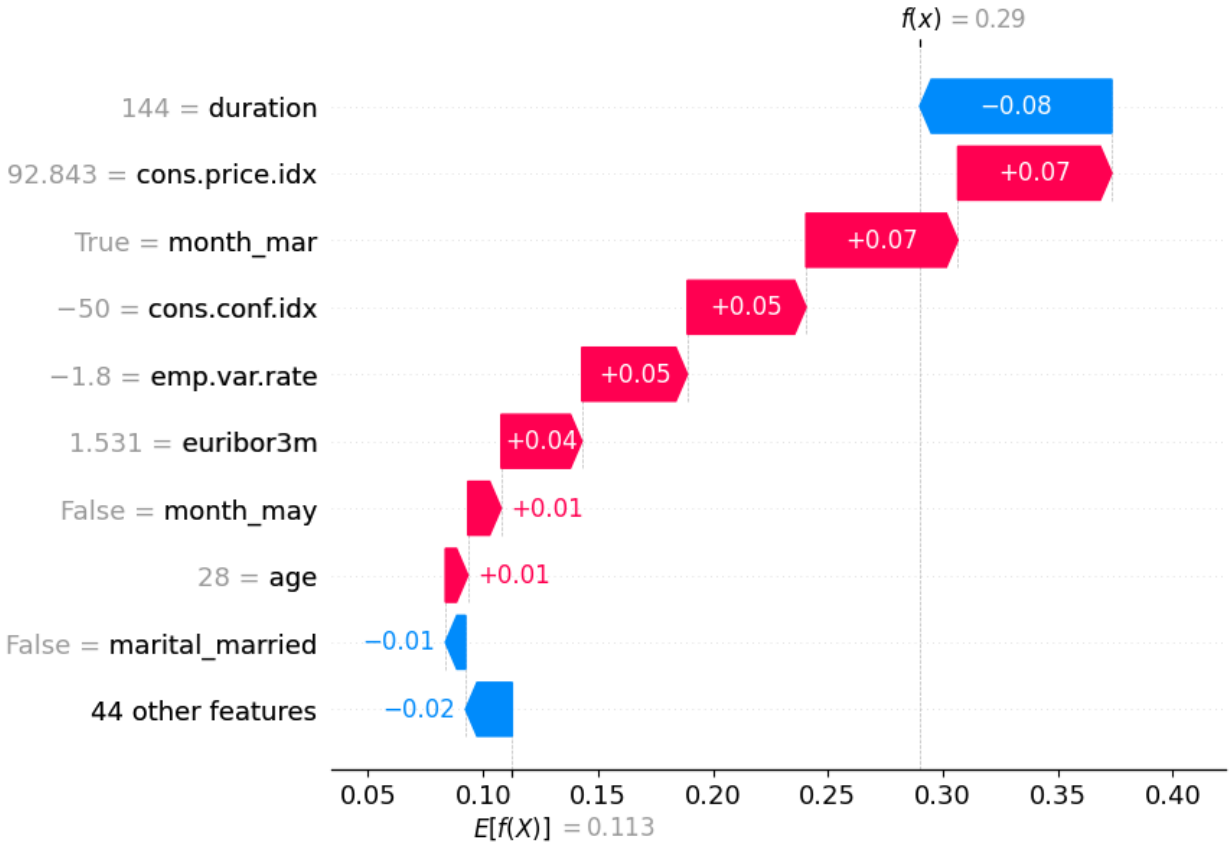

```
Customer Index 2 | Prediction: Will Not Subscribe (Prob: 0.30)
```




```
Customer Index 3 | Prediction: Will Not Subscribe (Prob: 0.00)
```



```
Customer Index 4 | Prediction: Will Not Subscribe (Prob: 0.29)
```



7. Conclusion and Insights

Based on the analysis and the SHAP values:

- **Duration:** The length of the last phone call is the most significant predictor. However, this information is only available after the call, making it less useful for predictive targeting before a campaign begins.
- **Economic Indicators:** Variables such as *euribor3m* (Euro Interbank Offered Rate)and *nr.employed* significantly impact the likelihood of subscription.
- **Recommendation:** The bank should prioritize customers who have been contacted fewer times in previous campaigns to reduce the risk of marketing fatigue and improve campaign effectiveness.